

Chinese dialectal parameterisation and microvariation: From Hong Kong to the world

As a child growing up in Hong Kong (HK) before its handover from Britain to China in July 1997, I grew up in a triglossic society where the three main language varieties in use were Cantonese, Mandarin and English (Kirkpatrick and Chau (2008), Bolton (2011)). In accordance with the education policy of ‘biliteracy and trilingualism’ (三言兩語) (Lee and Leung (2012)), I was brought up learning two languages (Chinese and English), and my native language (Chinese) consisted of two varieties (Cantonese and Mandarin), which were sociolinguistically aligned in ways that were typical of tri/diglossic societies (Ferguson (1959)). In many domains that are traditionally designated as H(igh), namely public discourse, written media and literary education, we used Mandarin, the national standard in the Chinese-speaking world. On the other hand, Cantonese, the local vernacular, was the default for most of the L(ow) domains, as we used Cantonese in our daily and social lives (Bauer (1984)).¹ Most educated and literate people of my generation possess bilingual proficiency in Chinese and English, and in addition to our native proficiency in Cantonese, we also possess written and reading proficiency in literary Mandarin, which puts us in good stead for learning how to speak and communicate in Mandarin. On a personal level, this multilingual environment shaped my interest and passion for languages, as the parallel uses of Cantonese and Mandarin (and, to a lesser extent, English) in sociolinguistically complementary (H/L) functions instilled a childhood obsession in me to make structural comparisons between these two forms of my native language. This also laid the foundation of my current work in comparative syntax² and teaching of Chinese (Mandarin and Cantonese) as a foreign language.

This paper outlines a formal parametric analysis of Chinese dialects, which is based on my research in Linguistics and underlies my method of teaching Chinese (Mandarin and Cantonese) as a foreign language. As my students in the past are either Western students who do not speak any Chinese or Chinese speakers who want to learn another dialect like Cantonese, a comparative method between Chinese dialects and other commonly spoken languages such as English and Romance has proven to be useful in introducing the grammar of Chinese to my students.^{3 4} This paper consists of three sections: section 1 presents the theoretical premises of formal comparative syntax that consist of the universal projection of functional categories and the parametric settings that underlie the systematic differences between different languages. Section 2 examines some well-known differences between Mandarin and Cantonese, which are presented as three types of parametric variation: 1) different properties of functional elements 2) different word order of the same lexical items 3) different constructions derived from parametric mechanisms that only occur in certain dialects. Section 3 concludes by providing a parametric

¹ In fact, HK Cantonese is more widely used than other varieties of Cantonese in their respective societies, as HK is the central hub of Cantonese media such as music (Cantopop) and films, which means that HK Cantonese is used in numerous traditionally H functions. Taking the sociolinguistic functions of H-Cantonese into account, Mandarin and English could be designated as super-H varieties that are used in relation to foreign visitors (cf Mkilifi’s (1972) analysis of Swahili-English bilingualism in Tanzania, which, like HK, is also a former British colony where English is imposed from above and Swahili is retained for many H-functions (Fasold (1984:45)).

² For a list of my presentations and publications, please visit my academic webpage:

<https://sites.google.com/alumni.york.ac.uk/kkt503> .

³ Despite claims of mutual unintelligibility due to phonological divergences, there is relative grammatical homogeneity between Chinese dialects and it is possible to reconstruct pan-Sinitic grammar that underlies most if not all Chinese dialects (Yue (2003)).

⁴ In addition to teaching grammar, I make corrections to my students’ pronunciation by introducing them to a consistent romanization system (jyutping) and demonstrating tone contours. For more advanced students, I also get them to practice using Chinese by listening to Cantonese videos and submitting written essays in Cantonese.

model of the Chinese verbal and nominal domains, which can enhance our understanding of different varieties of Chinese.

Section 1: Cartography and Parameters: Universal Projection of Functional Categories:

Formal comparative syntax assumes a universal cartography of functional categories on which language-specific parameters are set, and parametric variation is commonly derived from functional categories that can have different properties and hence underlie language variation (Borer (1983), Chomsky (1995), Biberauer (2008)). The two interrelated goals in parametric comparisons are 1) the reconstruction of the universal hierarchy of categories based on cross-linguistic comparisons 2) the parametric settings of individual languages, as determined by functional categories (Cinque and Rizzi (2010), Rizzi and Cinque (2016)). Cross-linguistic research has established versions of the following hierarchies for the clausal (1) and nominal (2) domains (Cinque (1999, 2002, 2006)):

- 1) C(omp) Subj(ect) T(ense) Adv(erb) V(erb)-Aff(ix) Obj(ect) S(entence) F(inal)P(article)
- 2) Possessor Poss Dem(onstrative) Num(ber) Q(uantifier) Cl(assifier) Adj(ective) N(oun)

As head-initiality is assumed for these two domains (Kayne (1994)), these cartographic structures (1-2) provide a neat representation of head-initial languages like Chinese, English and Romance, which facilitates Western speakers' acquisition of Chinese, as in all of these languages S(ubject)V(erb)O(bject) is the canonical word order (1) and determiners precede the noun (2). In mapping grammatical variation onto universal cartography, functional categories can be parameterised in terms of Merge, which, in current Minimalist thinking, consists of Internal Merge and External Merge (Chomsky (2004, 2008, 2013)), yielding the following types of functional categories (Roberts and Roussou (2003:209-213), Roberts (2001:97-107, 2007:267-282), Gianollo, Guardiano and Longobardi (2008:119), Roberts and Holmberg (2010:45-46)):

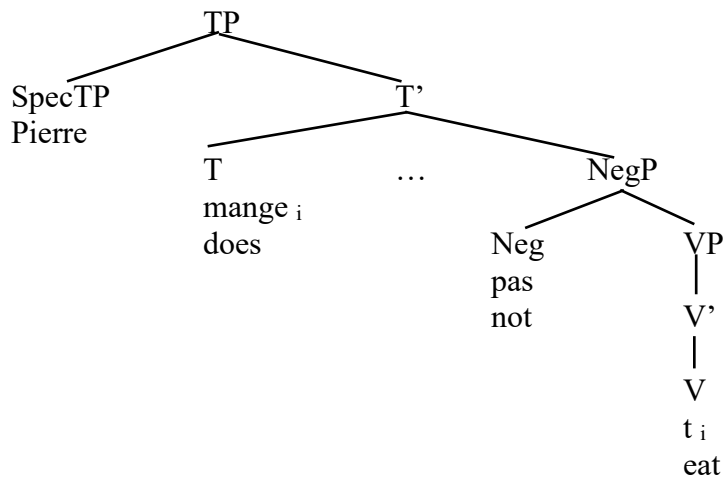
- 3) $F_{\text{MoveXP}} > F_{\text{MoveX}} > F_{\text{Agree}} > F_{\text{Merge}} > \emptyset$

Cartography and functional categories can hence be usefully applied to a systematic account of language variation, and in addition to the many different types of grammatical morphemes that lexicalise functional heads (F_{Merge}), differences in word order can be accounted for by functional categories that can either trigger movement ($F_{\text{MoveX/XP}}$) or not, in which case there is likely to be agreement (F_{Agree}) (Roberts (2010:50-51, 2012:353-354), cf Chomsky (2000, 2001)). In both the clausal and nominal domains, differences in the relative ordering of the lexical verb and noun can be accounted for by different properties of functional heads e.g. while in English lexical verbs and nouns are relatively low in their respective domains where they regularly come after all adverbs and adjectives, Romance languages regularly have lexical verbs and nouns come before certain adverbs and adjectives, as shown below:

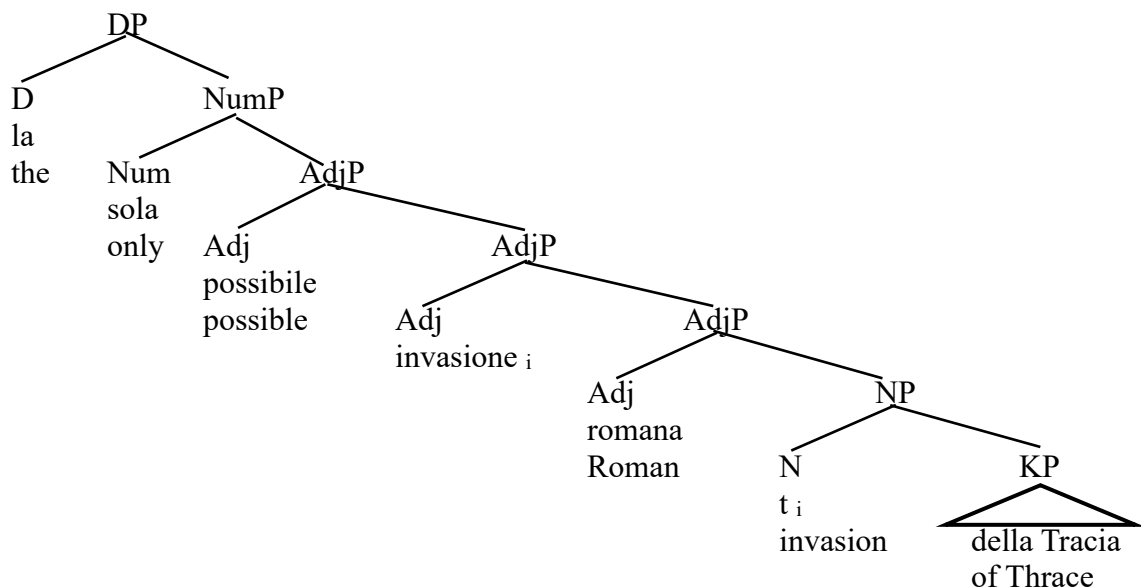
- 4a) Pierre ne mang-e pas
 Pierre NE eat-PRES.3SG NEG
 'Pierre does not eat.' (Pollock (1989:383-384, 393-395))
- 4b) la sola possible invasione romana de-lla Tracia
 The only possible invasion Roman of-the Thrace
 'The only possible Roman invasion of Thrace.' (Cinque (2010:1))

In the clausal domain that consists of a verb with a subject and an object and there are functional projections denoting Tense, Mood, Aspect and Negation (Belletti (1990), Cinque (1999)), it can be argued that in French the lexical verb (e.g. *mange* in 4a)) moves to a position above Neg (e.g. *pas* in 4a)) whereas English (e.g. *eat* in 4a)) does not. Instead, English resorts to *do*-support in Tense where it expresses subject agreement (e.g. *Pierre does not eat* in 4a)) (Pollock (1989:399-407)). Similarly, in the nominal domain which contains functional heads denoting definiteness, possession, quantification, and various types of adjectives (Abney (1987), Szabolcsi (1987), Ritter (1988, 1991), Lyons (1999), Longobardi (1994, 2001), Zamparelli (2005), Cinque (1993, 1994, 2010, 2014)), it can be argued that the lexical noun (e.g. *invasione* in 4b)) moves to an intermediate position between Num and Adj in Romance but not in English:

4a)



4b)



In terms of parametric settings, it can be argued that Romance languages such as French and Italian trigger movement (F_{MoveX}) of lexical verbs (4a) and nouns (4b), while in English these remain *in-situ* at the bottom of the tree where they enter into agreement (F_{Agree}) with their

functional heads. Based on this parametric model, we now proceed to examining the microvariation between Chinese dialects using Mandarin and Cantonese as our case study.

Section 2: Formal parametric comparison between Chinese dialects: Mandarin and Cantonese:

The previous section presented how parametric variation between different languages (e.g. English and Romance) are accounted for by syntactic operations (e.g. head movement of the lexical verb and noun) that are triggered by functional categories in some languages (Romance) but not in others (English) (4a-b). Based on this model of parametric variation, it is possible to posit three types of parametric variation between Cantonese and Mandarin, namely 1) grammatical forms corresponding to functional categories that have different selectional properties 2) different word order of the same lexical items 3) new constructions based on syntactic operations that occur in dialects but not in standard Mandarin. The first type is exemplified by Cantonese verbal suffixes, which are more numerous than in standard Mandarin and thus indicate finer distinctions in tense, aspectual and mood (TAM) properties (Matthews and Yip (2011:256)). As verbs undergo head movement for affixation (Huang (1982)), Chinese verbal affixes expressing TAM values can be modelled as functional heads triggering verb movement. Furthermore, even Cantonese affixes that have Mandarin equivalents are used more widely than in Mandarin, as some of them can be used with more types of verbs with different argument structure, which indicates a more generalised V-to-Asp movement that applies to all lexical verbs. The second type is seen in the well-known example of temporal adverb 先 ‘first, before’, which is preverbal in Mandarin but postverbal in Cantonese and other dialects (Sybesma (2013), Zhou (2018)). In fact, the same applies to the quantifiers 多 ‘more’ and 少 ‘less’, which are also preverbal in Mandarin but postverbal in Cantonese, and it is possible to parameterise adverbial placement of temporal and quantifying adverbs in Chinese dialects. The third type is exemplified by classifiers, which can be phrase-initial in Cantonese indicating definiteness and number (Cheng and Sybesma (1999), Heycock, Zamparelli, Cheng (2007)). This suggests that Cantonese CLs can function as articles, which can be derived by CL-to-D movement. Taken together, the microvariation between Cantonese and Mandarin constitutes a parametric model of the verbal and nominal domains that can be useful in learning (and teaching) Chinese.

Section 2.1: Cantonese verbal affixes: progressive V-住 *zyu*/緊 *gan*, possibility V-得 *dak*/到 *dou*, universal V-晒 *saai*, :

In comparison to Mandarin that has three commonly used verbal aspectual affixes (completive *-le* 了, stative *-zhe* 著, experiential *-guo* 過) (Sun (2006)) and a number of non-aspectual verbal affixes expressing deontic modality (*-ding* 定), possibility (*-dedao* 得到), reciprocity (*-hui* 回) and universal object relation (*-guang* 光), Cantonese has a richer inventory of verbal affixes that make more grammatical distinctions (Matthews and Yip (2011:225-226)). First of all, Cantonese has multiple verbal affixes expressing possibility and progressive aspect: while Mandarin V-(得)到(*de*)*dao* expresses possibility, Cantonese distinguishes V-得 *dak* (5b) and V-到 *dou* (5c), which express permission and possibility respectively:

- 5a) 我 說-(得)到 一定 做-(得)到
 Wo shuo-dedao yiding zuo-dedao
 I say-DEDAO must do-DEDAO
 ‘If I can say it, I can do it.’ (Mandarin) (Lyu (1980:151), in Yu (2022:11))

- 5b) 喂, 去-得 未 呀?
wai heoi-dak mei a
hey go-DAK NEG sentence-final.particle
'Hey, can we go yet?' (Cantonese) (Chappell and Peyraube (2016:307))
- 5c) 我哋 呢度 收-唔-到 大陸 台
ngodei nidou sau-ng-dou daailuk toi
we here receive-NEG-DOU mainland channel
'We cannot receive mainland channels here.' (Matthews and Yip (2011:251))

Similarly, while Mandarin uses the progressive verbal affix -著 *zhe* that often comes after the auxiliary (正)在(*zheng*)*zai* in expressing continuous aspect (6a), Cantonese has two progressive verbal affixes -住 *zyu* and -緊 *gan* that regularly come after the auxiliary 响度 *heongdou* (Matthews and Yip (2011:230-231)). However, Cantonese -住 *zyu* and -緊 *gan* Cantonese are not identical, since -緊 *gan* denotes continuous action and is regularly affixed onto dynamic verbs (6a), 住 *zyu* describes continuous state and is used with stative verbs (6b) (Matthews and Yip (2011:230-233)). It is hence possible to form minimal pairs where -住 *zyu* and -緊 *gan* denote the continuous state and action of the same verb (6c) e.g. 落雨 *lokjyu* 'to rain' where 落-緊雨 *lok-gan jyu* describes the ongoing action of raining and 落-住雨 *lok-zyu jyu* denotes the ongoing raining condition (Matthews and Yip (2011:225-227)):

- 6a) 嗰陣時 呢 棟 樓 仲 起-緊
gozansi ni dung lau zung hei-gan
then this CL building still build-PROG
'At the time this building was still in the process of being built.' (Cantonese) (Matthews and Yip (2011:231))
- 6b) 佢 成日 着-住 牛仔褲
Keoi sengjat zeok-zyu ngauzaifu
He always wear-PROG jeans
'He always wears jeans.' (Cantonese) (Matthews and Yip (2011:232))
- 6c) 外邊 响度 落-緊 雨 / 落-注 雨
Ngoimin heongdou lok-gan jyu lok-zyu jyu
Outside AUX drop-PROG rain drop-PROG rain
'It is raining outside.' (Cantonese) (Matthews and Yip (2011:225))⁵

⁵ The Mandarin equivalent would be 下-著雨 *xia-zhe yu* 'raining', which is closer to Cantonese 落-住雨 *lok-zyu jyu* as Mandarin does not have a verbal affix equivalent to Cantonese -緊 *gan*:

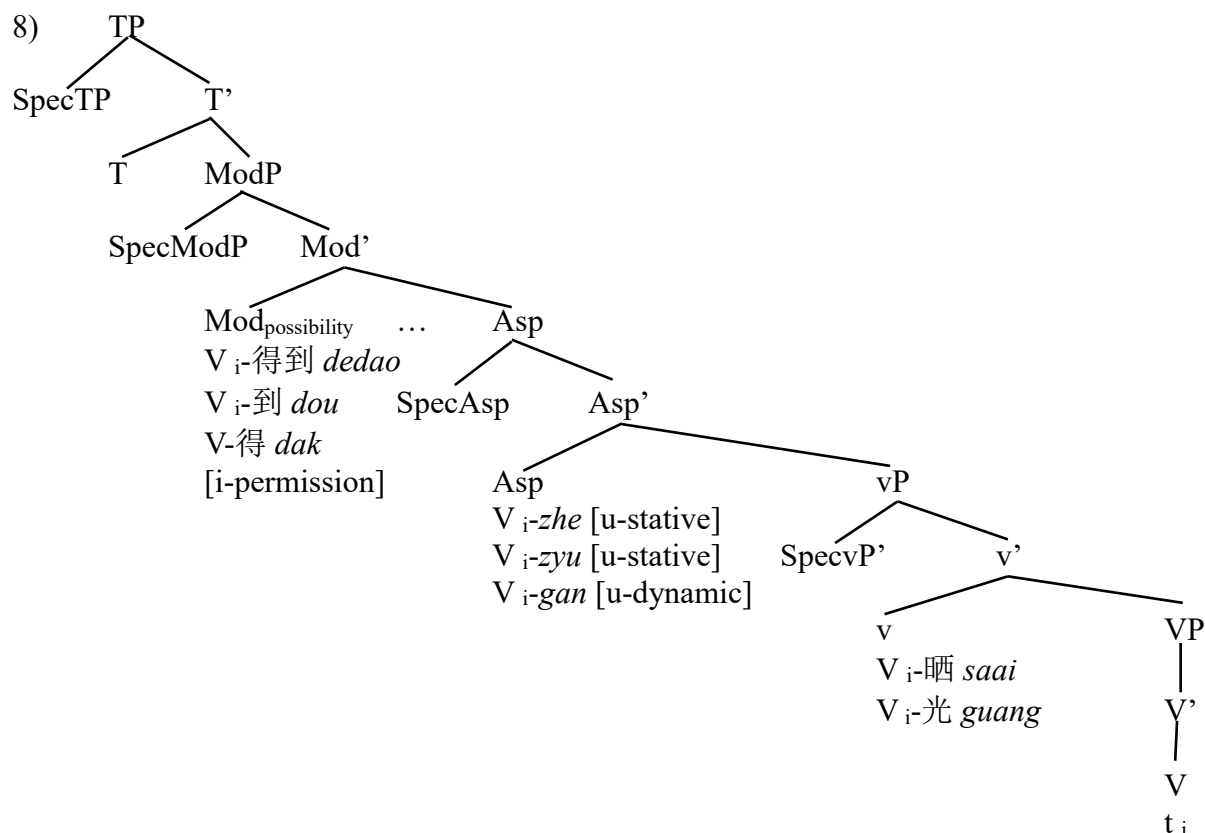
- i) 外面 正在 下-著 雨
waimian zhengzai xia-zhe yu
outside AUX drop-PROG rain
'It is raining outside.' (Mandarin)

More selectional differences are present in the verbal suffix that denotes universal argument application, which is V-光 *guang* in Mandarin and V-晒 *saai* in Cantonese. However, while Mandarin -光 *guang* is used mainly with one-place intransitive verbs (7b) and is marginally acceptable with two-place transitive verbs (7d), Cantonese V-晒 *saai* can be used with all types of verbs (7a, c) including three-place verbs (7e), which are incompatible with -光 in Mandarin (7f):

- 7a) 咁 人 走-晒 啦
 di jan zau-saai la
 the people leave-all sentence-final particle
 ‘The people all left.’ (Cantonese)
 (Matthews and Yip (2011:256), Yip and Matthews (2017:98))
- 7b) 人 走-光 了
 ren zou-guang le
 people leave-all sentence-final particle
 ‘The people all left.’ (Mandarin)
- 7c) 我 食-晒 咁 生果
 ngo sik-saai di saanggwo
 I eat-all the fruit
 ‘I have eaten all the fruits.’ (Cantonese) (Matthews and Yip (2011:256))
- 7d) (?)我 吃-光 水果
 wo chi-guang shuiguo
 I eat-all fruit
 ‘I have eaten all the fruits.’ (Mandarin)
- 7e) 我 俾-晒 咁 錢 佢哋
 ngo bei-saai di chin keoidei
 I give-all the money them
 ‘I have given all the money to them.’ OR ‘I have given all of them the money.’
 (Cantonese)
- 7f) *我 給-光 錢 他們
 wo gei-guang qian tamen
 I give-all money them
 ‘I have given all the money to them.’ (Mandarin)

The wider distribution of Cantonese verbal affixes indicates a verb phrase structure with more functional heads and more generalised verb movement: first of all, the distinction between V-得 *dak* (permission) and V-到 *dou* (possibility) suggests a difference in featural content in Mod_{Possibility} where V-得 *dak* denotes permission (5a-c). Secondly, the minimal pairing between 住 *zyu* and -緊 *gan* in denoting the progressive aspect of stative and dynamic verbs respectively (6a-c) indicates that the same functional head (Asp_{continuous}) can have different verbal selectional

properties. Moreover, the wider distribution of verb endings such as Cantonese V-晒 *saai* in comparison to Mandarin V-光 *guang* (7a-f) suggests a more generalised verb movement in the former that applies to lexical verbs, which can be represented as follows:



The next section examines adverbial placement as the same adverbs can occur in different positions in relation to the verb phrase, which indicates differences in word order and parameters in the verb phrase.

Section 2.2: Cantonese adverbs 先 *sin*/多 *do*/少 *siu*:

It is well known that Cantonese temporal adverb 先 *sin* ‘first, before’ is consistently postverbal (9a) when the same adverb 先 *xian* is preverbal in Mandarin (9b) (Sybesma (2013), Zhou (2018)), as shown below:

- 9a) 我哋 行 先
 ngodei haang sin
 we go first
 ‘Let’s go first.’ (Cantonese) (Zhou (2018:334))

- 9b) 我們 先 走
 womenxian zou

we first go
 ‘Let’s go first.’ (Mandarin)

The same can be said about quantifying adverbs 多 *do* ‘more’ (10a) and 少 *siu* ‘less’ (10c), which are also consistently preverbal in Mandarin (10b, d) but postverbal in Cantonese (10a, c):

10a) 我哋 諗-住 再 住 多 三 日
 ngodei nam-zyu zoi zyu do saam jat
 we think-ASP again live more three day
 ‘We are thinking of staying for three more days.’ (Cantonese)
 (Matthews and Yip (2010:315))

10b) 我們 想 再 多 住 三 天
 womenxiang zai duo zhu san tian
 we think again more live three day
 ‘We are thinking of staying for three more days.’ (Mandarin)

10c) 我 俾 少 十 分 嗰 個 學生
 ngo bei siu sap fan go go hoksaang
 I give less ten mark that CL student
 ‘I gave ten marks less to that student.’ (Cantonese) (Matthews and Yip (2010:315))

10d) 我 少 給 那 個 學生 十 分
 wo shao gei na ge xuesheng shi fen
 I less give that CL student ten mark
 ‘I gave ten marks less to that student.’ (Mandarin)

There are subtle differences between temporal adverb 先 *sin* ‘before’ and quantifying adverbs 多 *do*/少 *siu* ‘more/less’, since while the former comes after verb phrases including the object of the verb (11a), the latter two come immediately after the verb as they come between the verb and the object (11b-c), which indicates that 多 *do* and 少 *siu* are non-thematic complements of the verb (Li (1990:21-25)):

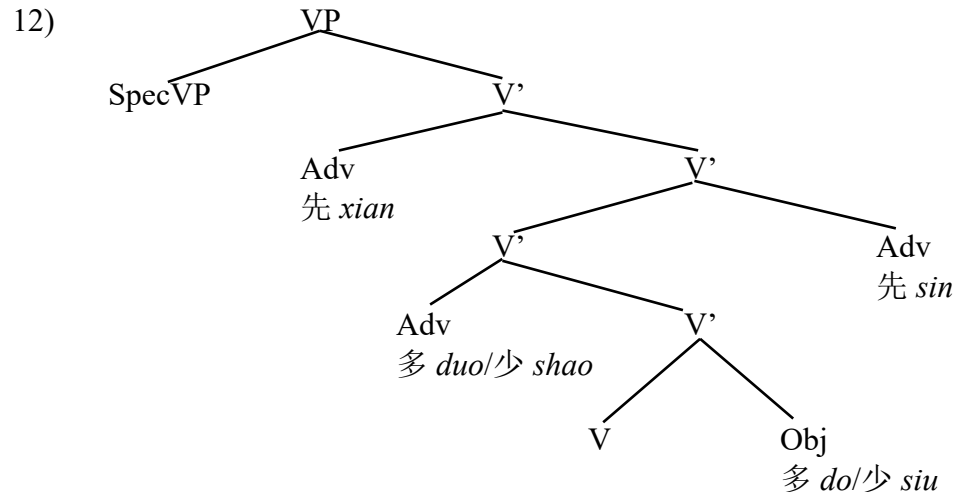
11a) 唔該 你 斟 杯 茶 俾 我 先
 nggoi nei zam bui cha bei ngo sin
 please you pour cup tea give me first
 ‘Please pour me a cup of tea first.’ (Cantonese) (Yuan (1983:228))

11b) 你 出 多 咗 條 問題
 nei chut do zo tiu mantai
 you give.out more PERF.ASP CL question
 ‘You gave out one question too many.’ (Yip and Matthews (2017:95))

11c) 公司 送 少 咗 一 樣 嘢 喇
 gungsi sung siu zo jat jeong je la

company deliver less PERF.ASP one CL thing sentence-final particle
 ‘The company delivered one item too few.’ (Yip and Matthews (2017:96))

The differences between temporal adverb 先 ‘before’ and quantifying adverbs 多 ‘more’/少 ‘less’ show that these are merged in different ways in the verb phrase: while 先 is a verbal adjunct that is merged at the V’-level either preverbally (9b) or postverbally (9a, 11a), 多 ‘more’ and 少 ‘less’ are either preverbal adverbs (10b, d) or postverbal complements (11b-c) where they are merged below V’, which can be represented as follows:



The next section examines constructions in Cantonese that are unattested in standard Mandarin, and these can be accounted for by syntactic operations that occur in the former but not in the latter, namely Classifier + NP where the classifier expresses referentiality and number and is equivalent to the article.

Section 2.3: Cantonese Classifiers: CL-D + NP:

A major difference between Mandarin and Cantonese nominal structures is that classifiers can head noun phrases in the latter but not in the former (Cheng and Sybesma (1999, 2005)), and these phrase-initial classifiers, like Western articles (e.g. English *the/a* and Romance *el/la/los/las* (Spanish)), show distinctions in definiteness and number, as shown in the examples below:⁶

13a) 隻 狗 鍾意 食 肉
 zek gau zungji sik juk
 CL.SG dog like eat meat
 ‘The dog likes to eat meat.’ (Cheng and Sybesma (1999:511))

13b) 啲 車 阻-住 個 出口
 di che zo-zyu go chuthau
 CL.PL car block-PROG CL exit

⁶ Such uses of the classifier are also attested in other dialects:

‘The cars are blocking the exit.’ (Cheng and Sybesma (1999:521))

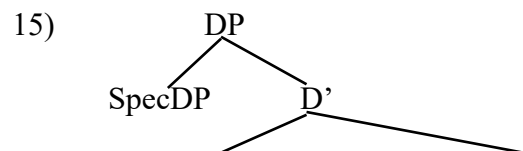
In contrast, standard Mandarin has a much more restricted distribution of CL + N which only occurs in object position where it denotes indefinite nouns (14a), and in order to express definite nouns, Mandarin resorts to either bare nouns (14b) or demonstratives that have their spatial relations weakened (14c) (Chen (2004)):

- 14a) 我 想 買 本 書
 wo xiang mai ben shu
 I want buy CL.SG book
 ‘I want to buy a book.’ (Cheng and Sybesma (1999:511))⁷

- 14b) 車 堵-著 出口
 che du-zhe chukou
 car block-ASP exit
 ‘The cars are blocking the exit.’ (Mandarin) (Cheng and Sybesma (1999:510))

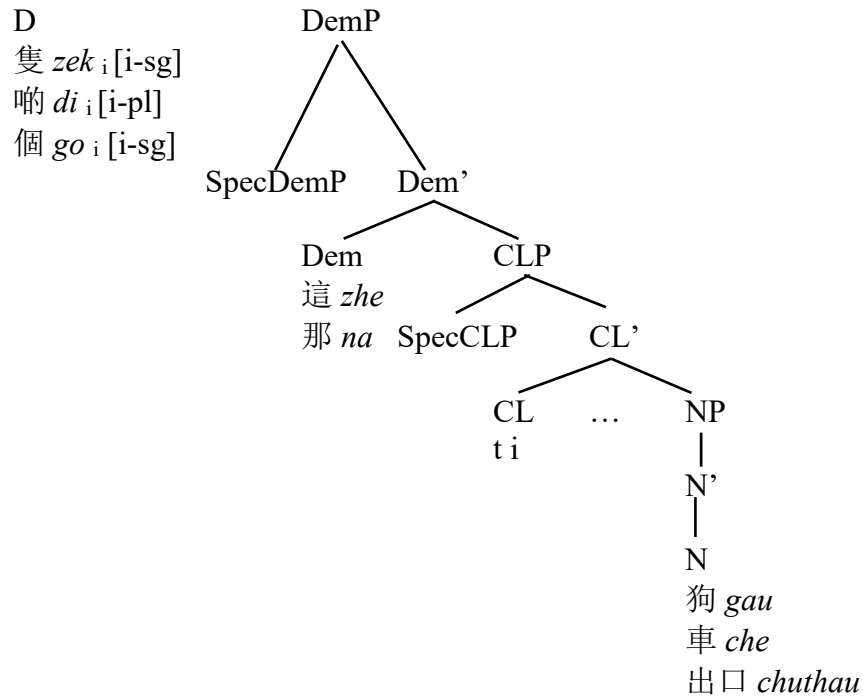
- 14c) 那 隻 狗 喜歡 吃 肉
 na zhi gou xihuan chi rou
 DEM CL.SG dog like eat meat
 ‘The dog likes to eat meat.’ (Mandarin) (Cheng, Heycock and Zamparelli (2007:80))

In accordance with the cartography of nominal elements where definiteness (D) is traditionally projected as a unique functional head at the top of the hierarchy (Lyons (1999), Longobardi (2001)), phrase-initial classifiers in Cantonese seem to be the result of CL-to-D movement which enables the classifier to function as definite articles that encode grammatical number, whereas in Mandarin the absence of CL-to-D movement is compensated by the use of demonstratives, as shown below:



⁷ In the Cantonese counterpart, the object here (CL + N) would be either definite or indefinite:

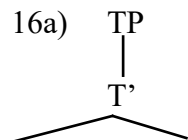
- i) 我 想 買 本 書
 ngo seong maai bun syu
 I want buy CL.SG book
 ‘I want to buy the book.’ OR ‘I want to buy a book.’ (Cantonese)
 (cf Cheng, Heycock and Zamparelli (2017:81))

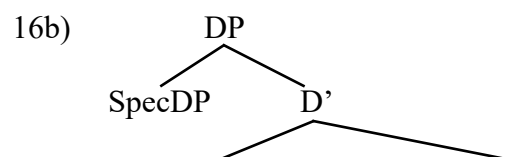
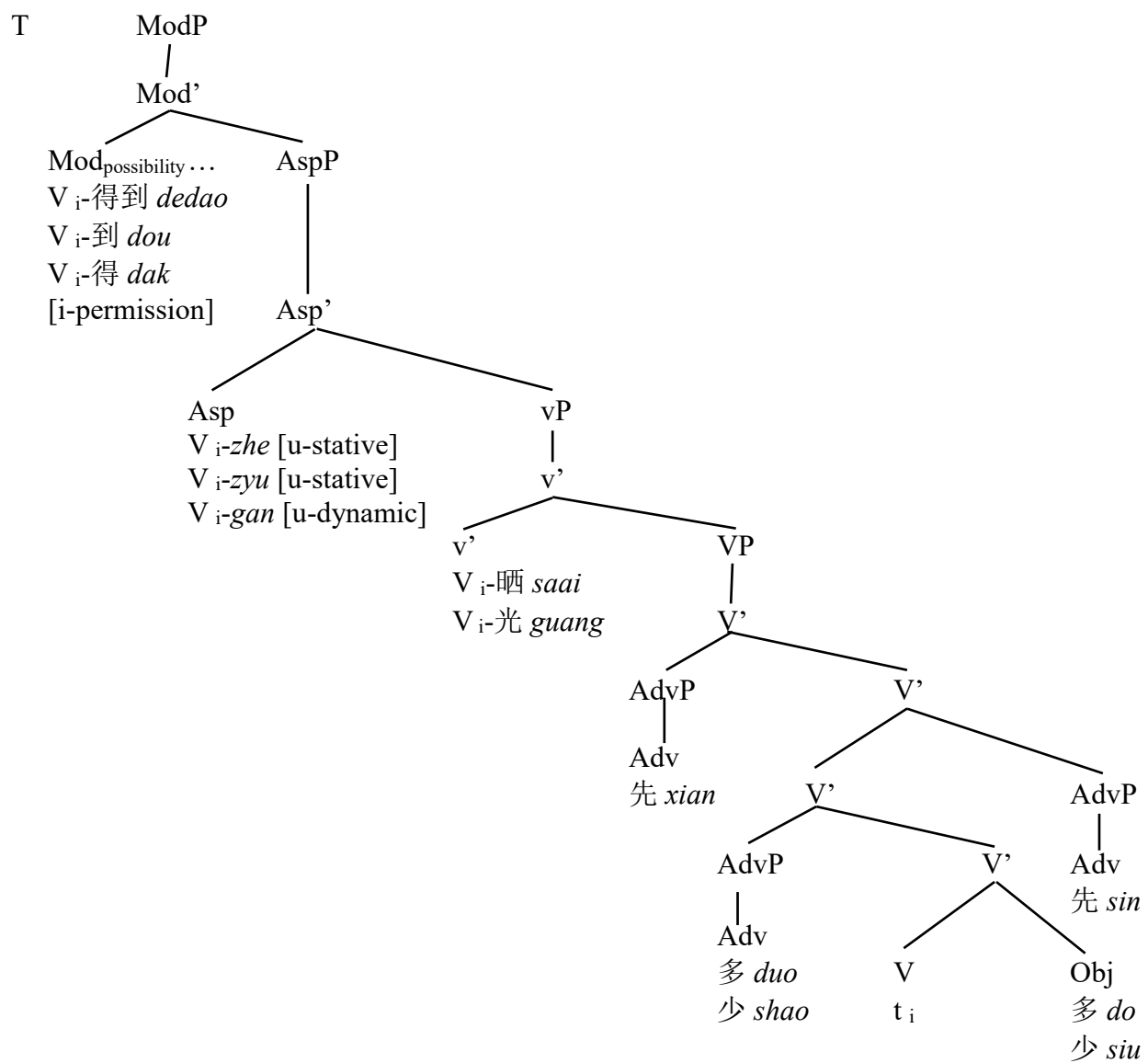


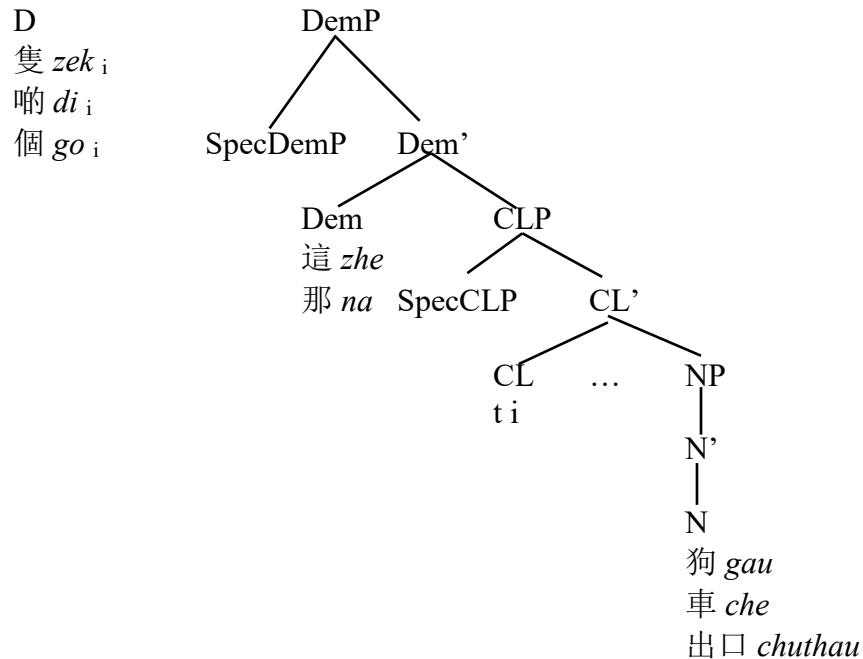
In the absence of the demonstratives (e.g. Mandarin 這 *zhe*/那 *na* in 14c)), the classifier can move to D and display distinctions in grammatical number, as seen in Cantonese (13a-b). Based on these empirical and formal differences, it is possible to parameterise the Chinese nominal and verbal domains, which will be presented in the next concluding section.

Section 3: Chinese Dialectal Parameterisation and Microvariation: Formal Parameters of Chinese Dialects (Mandarin and Cantonese):

The previous section illustrated three dimensions of parametric variation between Cantonese and Mandarin, which can be summarised as 1) different functional elements lexicalising different functional heads and having different selectional properties e.g. Cantonese V-得 *dak* (permission)/V-到 *dou* (possibility) vs Mandarin V-得到 *dedao* (possibility) (5a-c); Cantonese V-住 *zyu*/-緊 *gan* (Asp_{Progressive}) vs Mandarin V-著 *zhe* (Asp_{Progressive}) (6a-c); Cantonese V-晒 *saai* (all types of verbs, including ditransitives) vs Mandarin -光 *guang* (intransitives only) (7a-f) 2) different word order of the same lexical items e.g. temporal (先 'before') (9a-b) and quantifying adverbs (多 'more'/'少 'less') (10a-d) that can occur before or after the verb corresponding to different levels of the verb phrase (11a-c) 3) new syntactic operations, namely CL-to-D head movement in the nominal domain, which can account for phrase-initial classifiers in Cantonese (13a-b) which do not exist in Mandarin (14a-c). The parametric model of Chinese verbal (16a) and nominal (16b) domains, therefore, can be represented thus (cf 8, 12, 15):







This parametric model of Chinese dialects can significantly broaden our understanding of Chinese, as it allows us to recognise the microvariation that exists between dialects such as Mandarin and Cantonese. As is typical of people who grew up in di/triglossic societies and have acquired proficiency in more than one language variety, my native proficiency in Chinese is shared between standard Mandarin and the Cantonese vernacular, and dialectal microvariation is an inherent part of my native intuition.

Conclusion:

I consider myself fortunate to have been born and raised in a linguistically (and culturally) heterogeneous society like HK where I was exposed to a range of languages, which not only gave me multilingual proficiency but also structural sensitivity to closely related language varieties like Cantonese and Mandarin. My bidialectal proficiency, which underlies my comparison between these two Chinese dialects, is a natural consequence of my bilingual/triglossic upbringing in HK, and it is this ability that has allowed me to pursue my research in formal linguistics and language teaching. As the interest in Chinese grows and its research intensifies, it is necessary to recognise that Chinese is by no means a monolithic language in the form of standard Mandarin but one that is layered with sociolinguistic microvariation, which we can appreciate by applying parametric analysis to Chinese dialects.

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